RV College of Engineering®
Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Go, change the world

Optimization and Prediction of Fluoride Adsorption Using GTMAC-Modified Coconut Fiber via DOE and Hybrid Langmuir–Machine Learning Modeling

An Interdisciplinary Project Report (XX367P)

Submitted by,

Amritesh Sahu	**1RV23CD006**
Sai Likith Reddy	**1RV23CS212**
Aditi Amit Kallolkar	**1RV23CH002**
Hemanth H S	**1RV23CH019**

Under the guidance of

Dr. Jagadish H Patil
Associate Professor and Head
Dept. of CH
RV College of Engineering®

.

In partial fulfillment of the requirements for the degree of Bachelor of Engineering in respective departments

2025-26

RV College of Engineering®, Bengaluru

(*Autonomous institution affiliated to VTU, Belagavi*)

CERTIFICATE

Certified that the interdisciplinary project (XX367P) work titled ***Optimization and Prediction of Fluoride Adsorption Using GTMAC-Modified Coconut Fiber via DOE and Hybrid Langmuir–Machine Learning Modeling*** is carried out by of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in respective departments during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the interdisciplinary project report deposited in the departmental library. The interdisciplinary project report has been approved as it satisfies the academic requirements in respect of interdisciplinary project work prescribed by the institution for the said degree.

Dr. Jagadish H Patil Guide	Dr. Jagadish H Patil Head of the Department
Dr. M.V. Renukadevi Dean Academics	Dr. K. N. Subramanya Principal

External Viva

Name of Examiners	Signature with Date
1.	
2.	

DECLARATION

We, **Amritesh Sahu**, **Sai Likith Reddy**, **Aditi Amit Kallolkar**, **Hemanth H S** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**Optimization and Prediction of Fluoride Adsorption Using GTMAC-Modified Coconut Fiber via DOE and Hybrid Langmuir–Machine Learning Modeling**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

Name	Signature
1. Amritesh Sahu(1RV23CD006)	
2. Sai Likith Reddy(1RV23CS212)	
3. Aditi Amit Kallolkar(1RV23CH002)	
4. Hemanth H S(1RV23CH019)	

ACKNOWLEDGEMENTS

We are indebted to our guide, **Dr. Jagadish H Patil**, Associate Professor and Head, Department of CH, RVCE for the wholehearted support, suggestions and invaluable advice throughout our project work and also helped in the preparation of this thesis.

We also express our gratitude to our panel member **Dr. Manjula Sarode**, Assistant Professor, Department of CH, RVCE for theirvaluable comments and suggestions during the phase evaluations.

Our sincere thanks to the project coordinator **Dr. Vinutha Mosses**, Assistant Professor, Department of CH for timely instructions and support in coordinating the project.

Our gratitude to **Prof. Narashimaraja P**, Department of ECE, RVCE for the organized latex template which made report writing easy and interesting.

Our sincere thanks to **Dr. Jagadish H Patil**, Professor and Head, Department of Chemical Engineering, RVCE for the support and encouragement.

We are deeply grateful to **Dr. M.V. Renukadevi**, Professor and Dean Academics, RVCE, for the continuous support in the successful execution of this Interdisciplinary project.

We express sincere gratitude to our beloved Professor and Vice Principal, **Dr. Geetha K S**, RVCE and Principal, **Dr. K. N. Subramanya**, RVCE for the appreciation towards this project work.

Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.

ABSTRACT

Fluoride contamination in drinking water is a major environmental concern due to its harmful health effects. Conventional fluoride removal methods often involve high cost and complex processing, creating the need for low-cost and eco-friendly alternatives. In the present work, coconut fiber was selected as the base adsorbent material and surface modification using Glycidyl Trimethyl Ammonium Chloride (GTMAC) was carried out to improve fluoride adsorption characteristics.

The main objective of this work was to prepare and functionalize coconut fiber for fluoride adsorption studies and to develop a systematic experimental design using Design of Experiments (DOE). Biomaterial preparation involved drying, grinding, and sieving of coconut fiber, followed by GTMAC functionalization under alkaline conditions. Central Composite Design (CCD) was used for planning adsorption studies by varying parameters such as pH, adsorbent dose, fluoride concentration, and contact time.

Approximately 30 CCD experimental runs were generated using Design-Expert software for adsorption studies and optimization. Preliminary Langmuir adsorption model diagnostics carried out during Phase 2 showed acceptable prediction performance with an R^2 value of 0.8158 and RMSE value of 0.5278 mg/g, indicating good agreement between actual and predicted adsorption values.

CONTENTS

Abstract **i**

List of Figures **iv**

List of Tables **v**

Abbreviations **vi**

1 Fluoride Adsorption Optimization Using GTMAC-Modified Coconut Fiber and Machine Learning **1**

1.1 Introduction 2

1.2 Literature Review 3

1.2.1 Hybrid Langmuir–Machine Learning Modelling 4

1.3 Motivation 6

1.4 Problem statement 6

1.5 Objectives 6

1.6 Brief Methodology of the project 7

1.7 Assumptions made / Constraints of the project 7

2 THEORY AND FUNDAMENTALS OF FLUORIDE ADSORPTION USING GTMAC-MODIFIED COCONUT FIBER **8**

2.1 Fundamentals of Fluoride Adsorption 9

2.2 Surface Modification and GTMAC Functionalization 9

2.3 Batch Adsorption Studies 10

2.4 Design of Experiments (Design of Experiments (DOE)) and Langmuir Modelling 10

3 EXPERIMENTAL DESIGN AND PRE-ANALYSIS **12**

3.1 Specifications for the Design 13

3.2 Pre-analysis Work and Models Used 13

3.3 Experimental techniques 14

4 METHODOLOGY AND IMPLEMENTATION WORKFLOW 16

4.1 Methodology 17

5 Hybrid Langmuir–Machine Learning Modelling 20

5.1 Computational Modelling Workflow 21

5.2 Dataset and Input Factor Space 22

5.2.1 Experimental Factors and Their Significance 22

5.3 Physics-Based Response Generation 24

5.4 Langmuir Baseline Model 24

5.5 Residual Analysis 26

5.6 Feature Engineering for Machine Learning 27

5.7 Feature Importance and ML Readiness 28

6 Results & Discussions 29

6.1 Biomaterial Preparation and GTMAC Functionalization 30

6.2 Experimental Design and CCD Adsorption Runs 33

6.3 Langmuir Adsorption Model Diagnostics 34

7 Conclusion and Future Scope 36

7.1 Conclusion 37

7.2 Future Scope 37

7.3 Learning Outcomes of the Project 38

A Computational Workflow Scripts and Data Structures 39

A.1 First Appendix (Tcl Example) 40

A.2 Phase 1: Design of Experiments and Simulation 40

A.3 Phase 2: Langmuir Baseline Model Fitting 42

A.4 Phase 3: Feature Engineering and ML Preparation 43

A.5 Data File Structures (CSV Snippets) 43

Bibliography 45

RSV VIDYA SIKSHANA SAMITHI TRUST ® RV INSTITUTIONS

LIST OF FIGURES

1.1 Schematic representation of preparation, functionalization, and application of GTMAC-modified coconut fiber for fluoride removal from water. . 3

5.1 Diagnostic plots obtained from the Langmuir baseline model 25
5.2 Residual analysis and feature engineering diagnostic plots 27

6.1 Prepared bio material 31
6.2 Preparation of 10% NaOH solution 31
6.3 Addition of biomass to 10 %(w/v) NaOH solution 31
6.4 Biomass in 10% (w/v)NaOH solution 31
6.5 Washing of soaked biomass using distilled water 32
6.6 GTMAC modified material kept for drying 32
6.7 Graphs from the model built 35

RASHTREEYA SIKSHANA SAMITHI TRUST ® RV INSTITUTIONS

LIST OF TABLES

3.1 Experimental Parameters and Their Ranges 13

3.2 Instruments and Techniques Used 14

5.1 Hybrid modelling workflow followed in the project 21

5.2 Input factors used for the computational modelling dataset 23

5.3 Summary of generated adsorption response data 23

5.4 Performance of the Langmuir baseline model 25

5.5 pH-wise residual pattern observed after Langmuir fitting 26

5.6 Feature engineering groups used for residual learning 27

5.7 Top feature importance values from quick Random Forest residual model 28

6.1 CCD experimental runs and fluoride removal response 34

RASHTREEYA SIKSHANA SAMITHI TRUST
RV
INSTITUTIONS

ABBREVIATIONS

CCD Central Composite Design

CV Cross-Validation

DOE Design of Experiments

F^- ISE Fluoride Ion-Selective Electrode

FTIR Fourier Transform Infrared Spectroscopy

GTMAC Glycidyl Trimethyl Ammonium Chloride

LHS Latin Hypercube Sampling

MAE Mean Absolute Error

MLP Multilayer Perceptron

MSE Mean Squared Error

NOM Natural Organic Matter

OLS Ordinary Least Squares

PFO Pseudo-First-Order

PSO Pseudo-Second-Order

RF Random Forest

RMSE Root Mean Squared Error

RSM Response Surface Methodology

SEM Scanning Electron Microscopy

XGB Extreme Gradient Boosting

RAHTREEYA SIKSHANA SAMITHI TRUST ® RV INSTITUTIONS

Chapter 1

Fluoride Adsorption Optimization Using GTMAC-Modified Coconut Fiber and Machine Learning

CHAPTER 1

FLUORIDE ADSORPTION OPTIMIZATION USING GTMAC-MODIFIED COCONUT FIBER AND MACHINE LEARNING

Fluoride contamination in drinking water is a major environmental concern due to its harmful health effects. The need for low-cost and eco-friendly fluoride removal methods has increased interest in biomass-based adsorbents such as coconut fiber. In the present work, coconut fiber is modified using Glycidyl Trimethyl Ammonium Chloride (GTMAC) to improve fluoride adsorption characteristics. Design of Experiments (Design of Experiments (DOE)) using Central Composite Design (Central Composite Design (CCD)) is used for planning adsorption studies and optimization. This chapter presents the background, literature review, objectives, methodology, and organization of the report related to the study.

1.1 Introduction

These Fluoride contamination in drinking water is a major environmental and public health issue, especially in groundwater sources where fluoride levels may exceed the WHO permissible limit of 1.5 mg/L. Excess fluoride intake can cause dental and skeletal fluorosis, making effective defluoridation essential. Conventional treatment methods such as reverse osmosis, ion exchange, and activated alumina are efficient but often expensive and difficult to implement in small-scale or decentralized systems. Therefore, low-cost and sustainable biomass-based adsorbents have gained attention for water purification applications. Coconut fibre is an abundant agricultural waste material containing cellulose and lignin, but its natural fluoride adsorption capacity is limited due to the lack of sufficient active adsorption sites.

In this project, coconut fibre is chemically modified using Glycidyl Trimethyl Ammonium Chloride (GTMAC) to improve its fluoride adsorption performance by introducing positively charged quaternary ammonium groups on the surface. Batch adsorption studies are carried out using Design of Experiments (DOE) and Central Composite Design (CCD) to analyse the effect of important parameters such as pH, adsorbent dose, initial fluoride concentration, and contact time through multiple experimental runs. The study

also incorporates Langmuir isotherm modelling and machine learning-based prediction to enhance adsorption analysis and optimize fluoride removal efficiency. This integrated approach combines sustainable material development, statistical optimization, and predictive modelling for efficient fluoride removal from water.

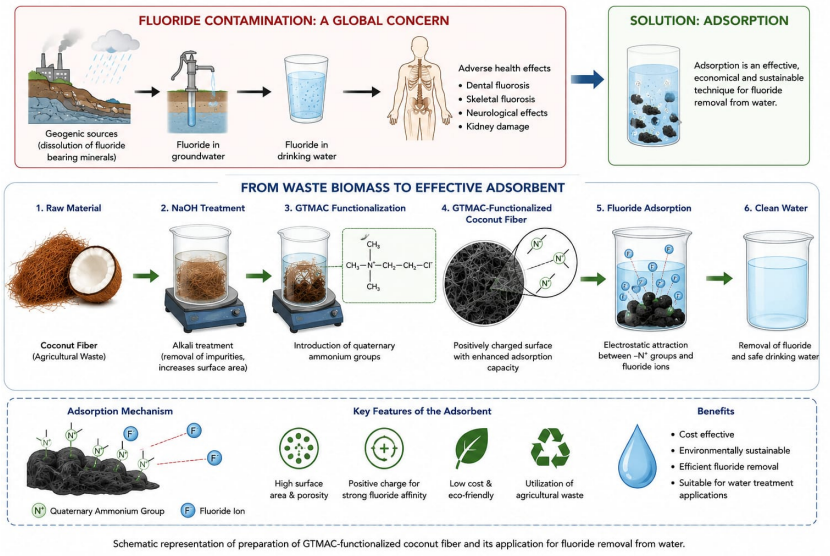


Figure 1.1: Schematic representation of preparation, functionalization, and application of GTMAC-modified coconut fiber for fluoride removal from water.

The schematic illustrates the overall approach adopted for fluoride removal from drinking water using GTMAC-functionalized coconut fiber as a bioadsorbent. The process begins with the utilization of coconut fiber, an agricultural waste biomass, which undergoes alkaline pretreatment using NaOH to remove impurities and activate the cellulose surface. The activated biomass is then functionalized using Glycidyl Trimethylammonium Chloride (GTMAC), introducing positively charged quaternary ammonium groups onto the surface. These functional groups enhance the affinity of the adsorbent toward negatively charged fluoride ions through electrostatic interaction and ion-exchange mechanisms. The modified adsorbent is subsequently applied in batch adsorption studies for fluoride removal from contaminated water. The figure also highlights the environmental and economic advantages of the proposed adsorbent, including low cost, sustainability, and utilization of waste biomass for water treatment applications.

1.2 Literature Review

Several researchers have explored the use of low-cost biomaterials for fluoride adsorption from water. Kamilia et al. (2025) reported that GTMAC-modified chitosan-

coated magnetic composites exhibited significantly improved adsorption capacity due to the introduction of quaternary ammonium groups [3]. Budeli et al. (2025) demonstrated that tri-metal oxide modified diatomaceous earth achieved nearly 98% fluoride removal efficiency over a wide pH range [2]. Ayalew (2023) studied thermally treated and acid-activated kaolin clay for fluoride adsorption and reported improved adsorption performance after activation [11]. Gogoi et al. (2019) showed that PEI-GTMAC modified biomass adsorbents exhibited enhanced anion adsorption capacity due to increased positive surface charge [14]. Similarly, Chen et al. (2014) explained that GTMAC quaternization increases the density of positively charged functional groups on biopolymers, thereby improving anion adsorption efficiency [21]. These studies confirm that chemical surface modification plays a major role in improving adsorption performance of biomass materials.

Researchers have also investigated adsorption modelling and optimization techniques for improving fluoride removal efficiency. Mohan et al. (2007) reported that activated alumina follows Langmuir adsorption behaviour and demonstrated strong fluoride affinity through ligand exchange mechanisms [27]. Mukherjee et al. (2023) used experimental modelling and parametric optimization for fluoride adsorption studies and highlighted the importance of statistical optimization methods in adsorption research [12]. Recent studies have also focused on integrating machine learning approaches with conventional adsorption models to improve prediction accuracy under complex operating conditions. The hybrid chemical-machine learning approach combines the physical understanding of adsorption obtained from Langmuir isotherm modelling with the predictive capability of machine learning algorithms for handling nonlinear interactions and residual errors. However, limited work has been reported on GTMAC-functionalized coconut fibre for fluoride removal using DOE optimization and hybrid predictive modelling [6]. Therefore, the present study focuses on developing a GTMAC-modified coconut fibre adsorbent, optimizing process parameters using Central Composite Design (CCD), and integrating Langmuir–Machine Learning modelling for improved fluoride adsorption prediction and optimization [9].

1.2.1 Hybrid Langmuir–Machine Learning Modelling

The integration of physics-based isotherm models with data-driven machine learning techniques has emerged as a powerful strategy for improving adsorption prediction accu-

racy beyond what either approach can achieve independently. The Langmuir isotherm provides a thermodynamically grounded description of monolayer adsorption on homogeneous sites, expressed through the equilibrium adsorption capacity q_e and the Langmuir constant K_L. However, in practice, real adsorption systems exhibit nonlinear interactions, heterogeneous surface effects, and parameter correlations that the Langmuir model alone cannot capture, leading to systematic prediction residuals.

Hybrid modelling frameworks that couple mechanistic models with machine learning have been reported across several adsorption and separation contexts. The core idea is to use the physics-based model to generate a baseline prediction and then train a machine learning algorithm to predict the residual error between the Langmuir prediction and the observed data. This residual correction strategy—sometimes called *grey-box* or *physics-informed machine learning*—allows the combined model to inherit the interpretability and extrapolation properties of the mechanistic model while exploiting the flexibility of data-driven methods to learn systematic deviations.

Several ensemble and neural network architectures have been evaluated for such residual correction tasks. Random Forest (Random Forest (RF)) regressors exploit bootstrap aggregation and feature importance ranking to capture variable interactions without over-fitting to small datasets. Extreme Gradient Boosting (Extreme Gradient Boosting (XGB)) offers superior handling of sparse feature spaces through sequential tree construction with regularization, making it well-suited to adsorption datasets with correlated input variables such as pH, dose, initial concentration, contact time, and temperature. Multilayer Perceptron (Multilayer Perceptron (MLP)) neural networks add the capacity to approximate arbitrary continuous functions, providing a complementary data-driven component when residual patterns are smooth and systematic.

In the context of fluoride adsorption, Latin Hypercube Sampling (Latin Hypercube Sampling (LHS)) has been proposed as an efficient design strategy for generating training datasets that span the full multi-dimensional factor space, reducing the number of simulations required compared to full-factorial designs while maintaining good space-filling properties. The hybrid workflow adopted in the present study follows this paradigm: Langmuir isotherm parameters are first estimated from experimental CCD data, synthetic training points are generated via LHS over the ten-dimensional factor space, Langmuir predictions are computed for each point, and the resulting residuals $\varepsilon = q_{\text{exp}} - q_{\text{Langmuir}}$

serve as the regression target for the machine learning models. The final hybrid prediction is:

$$q_{\text{hybrid}} = q_{\text{Langmuir}} + \hat{\varepsilon}_{\text{ML}} \tag{1.1}$$

where $\hat{\varepsilon}_{\text{ML}}$ is the residual predicted by the trained machine learning model. This formulation ensures that the hybrid model reduces to the Langmuir baseline when the machine learning correction is negligible, preserving physical interpretability. The approach is consistent with the broader trend in environmental engineering of combining first-principles knowledge with data-driven flexibility to achieve robust, generalizable predictions [8].

1.3 Motivation

Fluoride contamination in drinking water is a major health concern as excess fluoride can cause dental and skeletal fluorosis. Conventional fluoride removal methods are effective but expensive and difficult to implement in decentralized systems. This motivated the use of low-cost and eco-friendly biomass materials such as coconut fiber for adsorption studies. Since natural coconut fiber has limited fluoride adsorption capacity, GTMAC surface modification is used to improve its performance. The project also focuses on optimizing the adsorption process using DOE and improving prediction accuracy through hybrid Langmuir–Machine Learning modelling.

1.4 Problem statement

Coconut fiber is a low-cost biomass adsorbent but exhibits limited fluoride removal in its natural form. While GTMAC modification can enhance adsorption, the influence of interacting process parameters is not systematically optimized. Additionally, conventional models like the Langmuir isotherm have limited predictive accuracy under real conditions. This study addresses these gaps by combining surface modification, DOE-based optimization, and hybrid Langmuir–machine learning modeling to improve fluoride adsorption and prediction.

1.5 Objectives

The objectives of the project are

1. To apply Design of Experiments (DOE) to evaluate the influence and interaction of key process parameters and determine optimal conditions for fluoride removal.

2. To develop a surface-modified coconut fiber bio-adsorbent through GTMAC func-

tionalization for enhanced fluoride adsorption.

3. To enhance fluoride adsorption prediction by integrating the Langmuir isotherm model with machine learning-based error correction.

1.6 Brief Methodology of the project

The methodology adopted for the project involves preparation, surface modification, adsorption experimentation, optimization, and predictive modelling. Initially, raw coconut fiber is washed, dried, ground, and sieved to obtain uniform particle size. The prepared biomass is chemically activated using NaOH and further functionalized with Glycidyl Trimethyl Ammonium Chloride (GTMAC) to introduce positively charged quaternary ammonium groups on the surface. The modified adsorbent is then characterized using FTIR and SEM techniques to confirm surface modification and morphological changes. Batch adsorption studies are carried out to evaluate fluoride removal under different operating conditions such as pH, adsorbent dose, initial fluoride concentration, and contact time. Central Composite Design (CCD) under Response Surface Methodology (Response Surface Methodology (RSM)) is used to generate experimental runs and optimize the adsorption process. Finally, Langmuir isotherm modelling and machine learning-based error correction are used to improve adsorption prediction accuracy.

1.7 Assumptions made / Constraints of the project

The adsorption process is assumed to occur under controlled laboratory conditions.

Glycidyl Trimethyl Ammonium Chloride (GTMAC)-functionalized coconut fiber is assumed to have uniform surface properties.

Fluoride adsorption mainly occurs through electrostatic attraction and ion exchange.

Batch adsorption experiments are carried out with uniform mixing and constant temperature.

The Langmuir isotherm assumes monolayer adsorption on homogeneous sites.

The study is limited to the selected ranges of pH, adsorbent dose, concentration, and contact time used in CCD experiments.

Availability of characterization facilities such as Fourier Transform Infrared Spectroscopy (FTIR) and Scanning Electron Microscopy (SEM) may limit experimental analysis.

.

Chapter 2

THEORY AND FUNDAMENTALS OF FLUORIDE ADSORPTION USING GTMAC-MODIFIED COCONUT FIBER

CHAPTER 2

THEORY AND FUNDAMENTALS OF FLUORIDE ADSORPTION USING GTMAC-MODIFIED COCONUT FIBER

Fluoride adsorption using biomass-based adsorbents involves concepts from surface chemistry, environmental engineering, adsorption science, and statistical optimization. Understanding the fundamentals of adsorption mechanisms, surface modification techniques, adsorption isotherms, and experimental design methods is important for analyzing fluoride removal performance. This chapter discusses the theoretical background related to fluoride contamination, adsorption processes, GTMAC surface functionalization, batch adsorption studies, Langmuir isotherm modelling, and Design of Experiments (DOE). The chapter also explains the tools and techniques used for adsorption analysis and optimization in the present study.

2.1 Fundamentals of Fluoride Adsorption

Fluoride contamination in drinking water is a major environmental issue affecting both rural and urban populations. Excess fluoride concentration above the WHO permissible limit of 1.5 mg/L can cause dental fluorosis and skeletal fluorosis. Adsorption is one of the most widely used techniques for fluoride removal because of its simplicity, low cost, and high efficiency.

Adsorption is a surface phenomenon in which contaminants from a liquid phase accumulate on the surface of a solid material called an adsorbent. In the present work, GTMAC-modified coconut fiber acts as the adsorbent, while fluoride ions act as the adsorbate. The adsorption process mainly occurs through electrostatic attraction and ion-exchange mechanisms between fluoride ions and positively charged functional groups present on the modified adsorbent surface.

2.2 Surface Modification and GTMAC Functionalization

Natural coconut fiber contains cellulose, hemicellulose, and lignin with hydroxyl functional groups capable of adsorption. However, the natural adsorption capacity of coconut

fiber towards fluoride ions is limited due to low surface charge density and insufficient active adsorption sites. To improve adsorption efficiency, the coconut fiber is chemically modified using Glycidyl Trimethyl Ammonium Chloride (GTMAC). GTMAC introduces positively charged quaternary ammonium groups onto the adsorbent surface under alkaline conditions through etherification reactions.

The surface modification process involves washing, drying, grinding, and sieving of coconut fiber followed by NaOH pretreatment for activation of cellulose hydroxyl groups. The activated material is then reacted with GTMAC at controlled pH and temperature conditions. The introduced quaternary ammonium groups enhance fluoride adsorption through electrostatic attraction and ion-exchange mechanisms. FTIR and SEM characterization techniques are used to confirm successful surface modification and surface morphology changes in the prepared adsorbent.

2.3 Batch Adsorption Studies

Batch adsorption studies are performed to evaluate the fluoride removal efficiency of the prepared GTMAC-modified coconut fiber under different operating conditions. In batch adsorption experiments, a known amount of adsorbent is mixed with fluoride solution and agitated for a fixed period of time under controlled conditions.

The major process parameters affecting adsorption include pH, adsorbent dose, initial fluoride concentration, and contact time. The residual fluoride concentration is measured using a Fluoride Ion-Selective Electrode F^- ISE, and the adsorption performance is analyzed in terms of fluoride removal efficiency and adsorption capacity. Batch adsorption studies help in understanding adsorption equilibrium and the effect of different operating parameters on fluoride removal performance.

2.4 Design of Experiments (DOE) and Langmuir Modelling

The Design of Experiments (DOE) is a statistical technique used to systematically plan and analyze experiments involving multiple variables. In the present study, Central Composite Design (CCD) under Response Surface Methodology (RSM) is used to optimize fluoride adsorption conditions by studying the interaction between pH, adsorbent dose, fluoride concentration, and contact time.

The adsorption behaviour is further analyzed using the Langmuir adsorption isotherm

model, which assumes monolayer adsorption on homogeneous adsorption sites. The Langmuir equation is expressed as:

$$q_e = \frac{q_{max} b C_e}{1 + b C_e} \tag{2.1}$$

where q_e is the amount of fluoride adsorbed at equilibrium (mg/g), q_{max} is the maximum adsorption capacity (mg/g), b is the Langmuir constant related to adsorption energy (L/mg), and C_e is the equilibrium fluoride concentration in solution (mg/L).

The Langmuir model is used to study adsorption characteristics and adsorption capacity of the modified coconut fiber under different operating conditions. DOE and Langmuir modelling together help in understanding adsorption behavior and optimizing fluoride removal efficiency.

RASHTREEYA SIKSHANA SAMITHI TRUST ® RV INSTITUTIONS

Chapter 3

EXPERIMENTAL DESIGN AND PRE-ANALYSIS

CHAPTER 3

EXPERIMENTAL DESIGN AND PRE-ANALYSIS

Proper experimental planning is important for developing an efficient fluoride adsorption system. In the present work, GTMAC-modified coconut fiber is selected as the proposed adsorbent material for fluoride removal studies. Design of Experiments DOE using Central Composite Design CCD is adopted for planning the adsorption experiments and optimization studies. This chapter presents the experimental specifications, pre-analysis work, adsorption models, and experimental techniques selected for the study.

3.1 Specifications for the Design

The present study focuses on fluoride removal from aqueous solution using GTMAC-modified coconut fiber as the proposed adsorbent material. Batch adsorption studies are planned under controlled laboratory conditions by varying important adsorption parameters such as pH, adsorbent dose, fluoride concentration, and contact time. The selected operating ranges proposed for the adsorption studies are shown below.

Table 3.1: Experimental Parameters and Their Ranges

Parameter	**Range**
pH	2 – 7
Adsorbent Dose	0.1 – 1 g/100 mL
Initial Fluoride Concentration	2 – 70 ppm
Contact Time	30 – 180 min
Adsorbent	GTMAC-modified coconut fiber
Fluoride Measurement	Fluoride Ion-Selective Electrode
Experimental Design	Central Composite Design CCD

The adsorption experiments are proposed to be designed using Design-Expert software with approximately 30 CCD experimental runs for adsorption studies and optimization.

3.2 Pre-analysis Work and Models Used

Literature analysis was carried out to study previous research related to fluoride adsorption, biomass-based adsorbents, surface modification techniques, and adsorption modelling. Coconut fiber was selected as the proposed adsorbent because of its low cost, eco-friendly nature, and easy availability. Since raw coconut fiber has limited adsorption efficiency, Glycidyl Trimethyl Ammonium Chloride GTMAC was selected as the proposed surface modifying agent to improve fluoride adsorption through electrostatic attraction.

The adsorption behaviour is proposed to be analyzed using the Langmuir adsorption isotherm model, represented by: isotherm model, represented by:

$$q_e = \frac{q_{max} b C_e}{1 + b C_e} \tag{3.1}$$

where:

- q_e – adsorption capacity at equilibrium
- q_{max} – maximum adsorption capacity
- C_e – equilibrium fluoride concentration
- b – Langmuir adsorption constant

Design of Experiments (DOE) using Central Composite Design (CCD) was selected for planning adsorption studies and optimization. The adsorption data generated from the experiments are proposed to be analyzed using Design-Expert software.

The percentage fluoride removal is proposed to be calculated using:

$$\% \text{ Removal} = \frac{C_0 - C_e}{C_0} \times 100 \tag{3.2}$$

The adsorption capacity is proposed to be calculated using:

$$q_e = \frac{(C_0 - C_e)V}{m} \tag{3.3}$$

3.3 Experimental techniques

The following instruments and techniques were used during the experimental work:

Table 3.2: Instruments and Techniques Used

Experimental Stage	**Technique / Instrument**
Biomaterial Preparation	Hot air oven, grinder, sieve
Surface Modification	Magnetic stirrer, pH meter
GTMAC Functionalization	Sonication and alkaline treatment
Batch Adsorption Studies	Orbital shaker
Fluoride Measurement	Fluoride Ion-Selective Electrode
Surface Characterization	FTIR and SEM
Statistical Analysis	Design-Expert Software

This chapter presented the experimental specifications, adsorption models, pre-analysis work, and experimental techniques proposed for fluoride adsorption studies using GTMAC-modified coconut fiber. The selected adsorption parameters, DOE-based experimental design, and proposed analytical techniques discussed in this chapter form the basis for the proposed methodology and implementation workflow discussed in the next chapter.

RASHTREEYA SIKSHANA SAMITHI TRUST ® RV INSTITUTIONS

Chapter 4

METHODOLOGY AND IMPLEMENTATION WORKFLOW

CHAPTER 4

METHODOLOGY AND IMPLEMENTATION WORKFLOW

Proper implementation of adsorption studies is important for obtaining reliable fluoride removal data and adsorption analysis. The project involves preparation of GTMAC-modified coconut fiber, execution of batch adsorption experiments, and DOE-based adsorption studies under controlled laboratory conditions. Different adsorption parameters such as pH, adsorbent dose, fluoride concentration, and contact time are varied according to the CCD experimental matrix. This chapter presents the detailed methodology and implementation workflow followed for fluoride adsorption studies using GTMAC-functionalized coconut fiber.

4.1 Methodology

The methodology adopted for the present work consists of the following steps:

1. Collection of Raw Material: Raw coconut fiber is collected and used as the base adsorbent material for fluoride adsorption studies.

2. Washing and Drying of Biomass: The collected coconut fiber is washed thoroughly using distilled water to remove dust particles and impurities. The material is then dried in a hot air oven at approximately 80 °C to remove moisture content.

3. Grinding and Sieving: The dried coconut fiber is ground into fine powder using a mechanical grinder. Uniform particle size is obtained using a 125 BSS sieve for adsorption studies.

4. NaOH Pretreatment: The powdered biomass is treated using 10 percent NaOH solution for alkaline activation. The pretreatment process removes lignin and impurities while exposing cellulose hydroxyl groups required for surface modification. The treated biomass is washed repeatedly until neutral pH is obtained.

5. GTMAC Functionalization: Surface modification is carried out using Glycidyl Trimethyl Ammonium Chloride (GTMAC) under alkaline conditions. The reaction is maintained at approximately 60 °C for 5 hours with sonication. GTMAC

introduces positively charged quaternary ammonium groups onto the adsorbent surface, improving fluoride adsorption through electrostatic attraction.

6. Drying and Storage of Modified Adsorbent: After functionalization, the modified adsorbent is washed thoroughly using distilled water and dried in a hot air oven. The prepared adsorbent is stored in airtight containers for adsorption studies.

7. Characterization Studies: Fourier Transform Infrared Spectroscopy (FTIR) is used to identify functional groups present on the adsorbent surface and confirm GTMAC modification. Scanning Electron Microscopy (SEM) is used to study the surface morphology and structural changes after chemical modification.

8. Preparation of Fluoride Solution: Fluoride stock solution is prepared using distilled water. Different fluoride concentrations are prepared according to the CCD experimental design.

9. Batch Adsorption Studies: A fixed quantity of GTMAC-modified coconut fiber is added to fluoride solution under controlled experimental conditions. The adsorption studies are conducted by varying:

    - pH
    - Adsorbent dose
    - Initial fluoride concentration
    - Contact time

10. Agitation and Adsorption Process: The adsorption mixtures are agitated using an orbital shaker for a specified contact time to ensure proper interaction between fluoride ions and the adsorbent surface.

11. Filtration and Fluoride Measurement: After adsorption, the solution is filtered and residual fluoride concentration is measured using a Fluoride Ion-Selective Electrode (F^- ISE).

12. CCD Experimental Runs: Approximately 30 experimental runs are conducted according to the Central Composite Design (CCD) matrix generated using Design-Expert software.

13. **Adsorption Calculations:** The percentage fluoride removal is calculated using:

$$\%\ Removal = \frac{C_0 - C_e}{C_0} \times 100 \tag{4.1}$$

The adsorption capacity is calculated using:

$$q_e = \frac{(C_0 - C_e)V}{m} \tag{4.2}$$

where:

- C_0 = Initial fluoride concentration (mg/L)
- C_e = Equilibrium fluoride concentration (mg/L)
- V = Volume of solution (L)
- m = Mass of adsorbent used (g)
- q_e = Adsorption capacity (mg/g)

14. **Data Analysis and Optimization:** The adsorption data obtained from all experimental runs are tabulated and analyzed using Design-Expert software for adsorption response analysis and optimization studies.

This chapter presented the detailed methodology and implementation workflow followed for fluoride adsorption studies using GTMAC-modified coconut fiber. The preparation of adsorbent, GTMAC functionalization process, characterization studies, batch adsorption experiments, CCD experimental runs, and adsorption analysis procedure were discussed in detail. The adsorption results and optimization analysis obtained from these studies are presented in the next chapter.

Chapter 5

Hybrid Langmuir–Machine Learning Modelling

CHAPTER 5

HYBRID LANGMUIR–MACHINE LEARNING MODELLING

The present project includes a computational modelling workflow for improving fluoride adsorption prediction using a hybrid Langmuir–Machine Learning approach. The purpose of this modelling stage is to combine the interpretability of adsorption science with the predictive capability of data-driven residual correction. The Langmuir model is used as the baseline adsorption model, and machine learning is used to learn the remaining nonlinear error patterns that are not captured by the baseline model.

The modelling work was organized into three completed phases. In Phase 1, a 500-sample design matrix was generated using Latin Hypercube Sampling across ten adsorption-related factors. A physics-based simulation was then used to generate adsorption capacity values. In Phase 2, a multi-factor Langmuir baseline model was fitted to the simulated adsorption response. In Phase 3, residual analysis and feature engineering were carried out to prepare machine-learning-ready training and testing datasets. This chapter presents the modelling data, algorithmic workflow, residual analysis, and machine learning readiness obtained from the project files.

5.1 Computational Modelling Workflow

The hybrid modelling approach follows a sequential workflow. First, the adsorption factor space is sampled to generate a broad set of operating conditions. Then, fluoride adsorption capacity is estimated using a physics-based response model. The Langmuir baseline model is fitted to this dataset and evaluated using standard regression metrics. The residuals from this model are then analyzed to identify systematic patterns. Finally, engineered features are generated so that machine learning models can learn the residual correction.

Table 5.1: Hybrid modelling workflow followed in the project

Phase	Task	Main Input	Main Output
1	LHS experimental design	Ten selected adsorption factors	500 design points
1	Physics-based response simulation	LHS design matrix	Simulated `q_removal` dataset
2	Langmuir baseline fitting	Simulated adsorption dataset	Predicted values and residuals
3	Residual analysis	Langmuir residuals	Dominant residual patterns
3	Feature engineering	Original factors and residuals	ML training and testing datasets

The final hybrid prediction is intended to be calculated by adding the machine-learning residual correction to the Langmuir prediction, as shown below:

$$q_{hybrid} = q_{Langmuir} + \hat{e}_{ML} \tag{5.1}$$

where q_{hybrid} is the final predicted adsorption capacity, $q_{Langmuir}$ is the baseline Langmuir prediction, and $\hat{e}_{ML}$ is the residual correction predicted by the machine learning model.

5.2 Dataset and Input Factor Space

The modelling dataset was generated using Latin Hypercube Sampling (LHS), which provides uniform coverage of the selected factor space. A total of 500 samples were generated across ten input factors. The factors include major adsorption parameters such as pH, initial fluoride concentration, contact time, adsorbent dose, temperature, and additional water-quality factors such as chloride, hardness, carbonate, and natural organic matter.

5.2.1 Experimental Factors and Their Significance

The selection and scientific justification for each of the ten factors are outlined below:

- **pH (3.01–8.99):** The most critical factor; it controls the surface charge of the adsorbent and fluoride speciation. Optimal removal occurs around pH 6.5.
- **Initial Fluoride Concentration (1.01–9.98 mg/L):** Drives the Langmuir equilibrium and represents typical groundwater contamination levels.
- **Contact Time (10–120 min):** Governs the pseudo-second-order kinetics required to reach adsorption equilibrium.
- **Adsorbent Dose (0.51–5.00 g/L):** Determines the total availability of active adsorption sites.
- **Temperature (20.0–40.0 °C):** Influences the adsorption capacity through Arrhenius temperature-dependent kinetics.
- **Flow Rate (0.50–2.00 L/min):** Determines the hydraulic residence time in continuous treatment systems.

- **Chloride (0–100 mg/L):** Acts as a competing anion in the water matrix.
- **Hardness (1–499 mg/L as $CaCO_3$):** Represents cation competition (Ca^{2+}/Mg^{2+}) for available surface sites.
- **Carbonate (0–100 mg/L as HCO_3^-):** A strong competing anion whose effect is highly dependent on pH speciation.
- **Natural Organic Matter (0–50 mg/L):** Causes pore fouling, which physically blocks access to the adsorbent's internal surface area.

Table 5.2: Input factors used for the computational modelling dataset

Sl. No.	Factor	Minimum	Maximum	Unit
1	pH	3.01	8.99	–
2	Initial fluoride concentration, C_0	1.01	9.98	mg/L
3	Contact time	10	120	min
4	Adsorbent dose	0.51	5.00	g/L
5	Temperature	20.0	40.0	°C
6	Flow rate	0.502	1.999	L/min
7	Chloride concentration	0	100	mg/L
8	Hardness	1	499	mg/L as $CaCO_3$
9	Carbonate concentration	0	100	mg/L as HCO_3^-
10	Natural organic matter	0	50	mg/L

The generated dataset contains 500 rows and 13 columns, including run number, input factors, randomization order, and adsorption response. The simulated adsorption capacity `q_removal` varied from 1.4166 mg/g to 8.3156 mg/g, with a mean value of 4.0989 mg/g and a standard deviation of 1.2309 mg/g. These values indicate a physically reasonable adsorption capacity range for the modelling study.

Table 5.3: Summary of generated adsorption response data

Quantity	Value
Total samples	500
Number of input factors	10
Minimum `q_removal`	1.4166 mg/g
Maximum `q_removal`	8.3156 mg/g
Mean `q_removal`	4.0989 mg/g
Standard deviation	1.2309 mg/g

5.3 Physics-Based Response Generation

The response generation script was developed to simulate fluoride adsorption capacity under different operating conditions. The simulation considered important adsorption mechanisms such as pH-dependent adsorption, concentration effect, contact-time response, temperature effect, dose saturation, flow-rate effect, ion competition, and natural organic matter fouling. The reference Langmuir parameters used in the simulation were $q_{max} = 8.5$ mg/g and $K_L = 0.12$ L/mg at 25 °C. The optimum pH was taken as 6.5, and a bell-curve relationship was used to represent pH-dependent adsorption behaviour.

The pH effect was modelled using:

$$f_{pH} = \exp\left[-\frac{(pH - 6.5)^2}{2(1.1)^2}\right] \tag{5.2}$$

This relationship gives maximum adsorption near pH 6.5 and lower adsorption at strongly acidic or alkaline conditions. Ion competition and natural organic matter effects were included as moderate reduction factors so that the simulated response remained physically realistic. A small measurement noise of 5% was also added to represent experimental variability.

5.4 Langmuir Baseline Model

The Phase 2 model used the simulated adsorption dataset to fit a multi-factor Langmuir baseline model. The ten original factors were standardized and expanded into second-order polynomial features. This produced 66 model features, including original terms, squared terms, and two-way interaction terms. A linear regression model was then fitted on the expanded feature matrix to estimate adsorption capacity.

The residual error was calculated as:

$$e = q_{actual} - q_{predicted} \tag{5.3}$$

where q_{actual} is the simulated adsorption capacity and $q_{predicted}$ is the Langmuir baseline prediction.

Table 5.4: Performance of the Langmuir baseline model

Metric	**Value**
Number of samples	500
Original factors	10
Expanded polynomial features	66
Model type	Multi-factor Langmuir polynomial model
R^2	0.8158
RMSE	0.5278 mg/g
MAE	0.4282 mg/g
Residual mean	-6.63×10^{-16}
Residual standard deviation	0.5278 mg/g

The Langmuir baseline model explained 81.58% of the variation in adsorption capacity. The near-zero residual mean indicates that the model was unbiased overall. However, the remaining residual variation showed systematic behaviour, which supports the use of machine learning for residual correction.

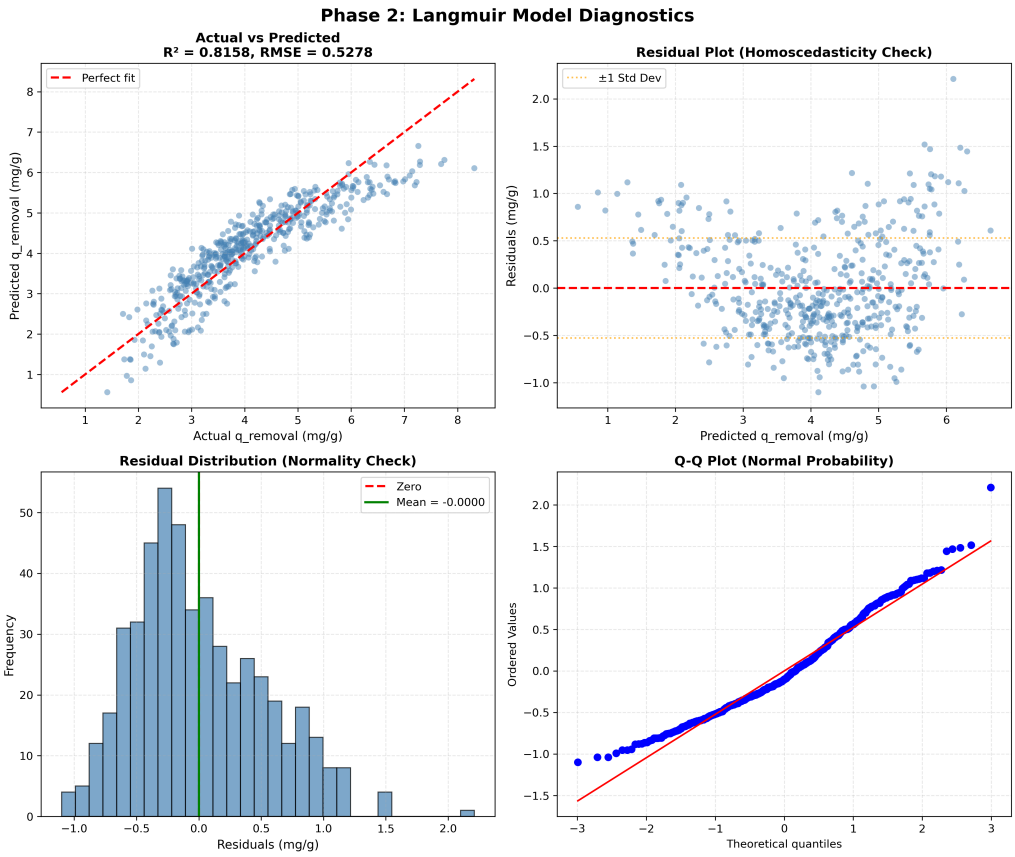


Figure 5.1: Diagnostic plots obtained from the Langmuir baseline model

5.5 Residual Analysis

Residual analysis was carried out to identify the patterns not captured by the Langmuir baseline model. The residuals ranged from -1.1008 mg/g to 2.2100 mg/g. The strongest pattern was observed with pH. The model underestimated adsorption near the optimum pH range and overestimated adsorption in some acidic regions. This indicates that the polynomial approximation could not completely capture the bell-shaped pH behaviour introduced during response generation.

Table 5.5: pH-wise residual pattern observed after Langmuir fitting

pH range	**Number of points**	**Mean residual (mg/g)**	**Interpretation**
pH 3–4	84	0.2478	Underestimation
pH 4–5	83	-0.5250	Overestimation
pH 5–6	84	-0.1172	Mild overestimation
pH 6–6.5	41	0.4706	Underestimation
pH 6.5–7	42	0.6491	Strong underestimation
pH 7–7.5	41	0.2082	Underestimation
pH 7.5–8	42	-0.1760	Overestimation
pH 8–9	83	-0.1819	Overestimation

The most important residual finding was observed in the pH 6.5–7 range, where the mean residual was +0.6491 mg/g. A positive residual means that the baseline model predicted a lower adsorption value than the actual response. Therefore, this region provides an important opportunity for machine learning correction.

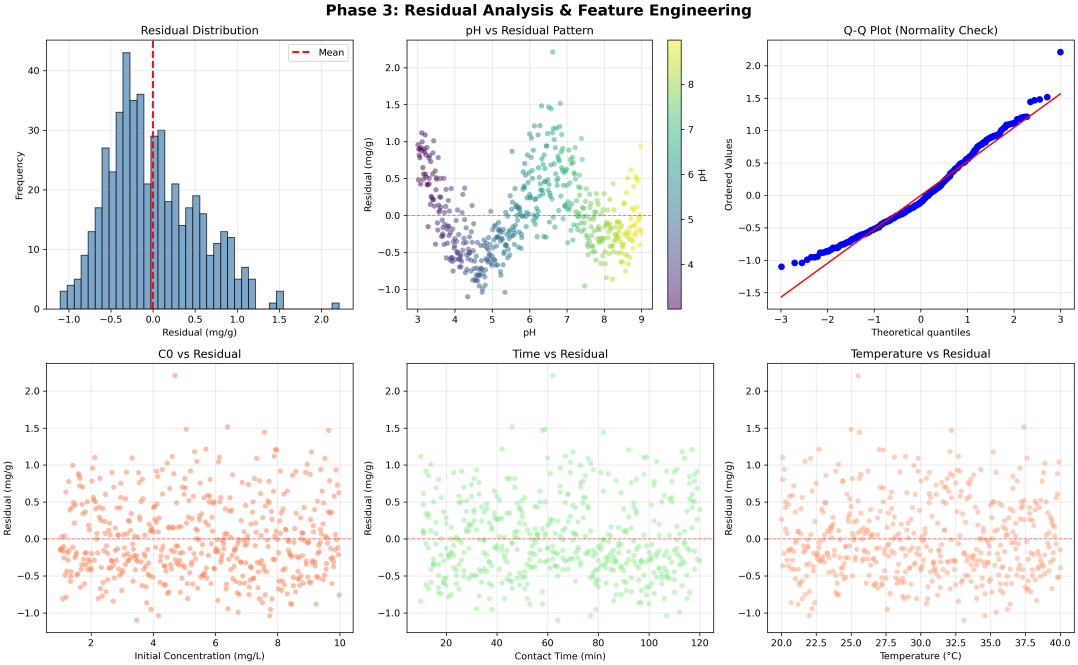


Figure 5.2: Residual analysis and feature engineering diagnostic plots

5.6 Feature Engineering for Machine Learning

Feature engineering was carried out to convert the original factor data into more informative machine learning inputs. The original ten factors were retained, and 28 new features were generated. Therefore, the final feature matrix contained 38 input features. The engineered features were grouped into pH-deviation features, ion-competition features, equilibrium and loading features, logarithmic transformations, interaction terms, and composite performance indicators.

Table 5.6: Feature engineering groups used for residual learning

Feature group	**Number of features**	**Purpose**
Original adsorption factors	10	Preserve direct experimental variables
pH deviation features	5	Capture distance from optimum pH 6.5
Ion competition features	5	Represent combined effects of chloride, hardness, and carbonate
Equilibrium and loading features	5	Represent dose, concentration, time, and fouling effects
Logarithmic transforms	5	Capture nonlinear scale effects
Interaction terms	5	Represent coupled factor effects
Composite indicators	3	Represent combined performance and stress conditions
Total input features	38	Machine learning feature matrix

The final dataset was divided into training and testing subsets. The training dataset contained 400 samples, and the testing dataset contained 100 samples. Both datasets contained 38 input features and one residual target column. This prepared dataset is

suitable for training residual correction models such as Random Forest, XGBoost, and Multi-Layer Perceptron.

5.7 Feature Importance and ML Readiness

A quick Random Forest model was trained during Phase 3 to verify whether the residual pattern was learnable. The model achieved a training R^2 of 0.9039 and a testing R^2 of 0.4729 on residual prediction. This result shows that the residuals are not random and contain learnable structure. The feature importance ranking also confirmed that pH-related features dominate the residual correction problem.

Table 5.7: Top feature importance values from quick Random Forest residual model

Rank	Feature	Importance
1	pH_abs_dev	0.2109
2	pH_dev_sq	0.1669
3	performance_index	0.0787
4	optimal_pH_score	0.0523
5	pH	0.0498
6	pH_dev	0.0478
7	pH_optimal	0.0454
8	pH_gaussian	0.0440

The top two features, pH_abs_dev and pH_dev_sq, together contributed strongly to residual prediction. This confirms that the baseline Langmuir polynomial model captured the broad adsorption behaviour but did not fully represent the nonlinear pH optimum. The prepared Phase 3 outputs therefore provide the required foundation for Phase 4 machine learning model training and Phase 5 hybrid model integration.

This chapter presented the complete modelling information extracted from the project files, including dataset generation, Langmuir baseline fitting, residual analysis, feature engineering, and machine learning readiness. The experimental results, biomaterial preparation observations, CCD adsorption runs, and Langmuir diagnostic outcomes are discussed in the next chapter.

Chapter 6

Results & Discussions

CHAPTER 6

RESULTS & DISCUSSIONS

he present work focused on the preparation and functionalization of coconut fiber for fluoride adsorption studies. GTMAC-modified coconut fiber was prepared and adsorption experiments were planned using Design of Experiments (DOE) through Central Composite Design (CCD). Preliminary Langmuir adsorption model diagnostics were also carried out during Phase 2 of the study. This chapter presents the results obtained during biomaterial preparation, GTMAC functionalization, CCD experimental design, and Langmuir adsorption model analysis.

6.1 Biomaterial Preparation and GTMAC Functionalization

Aaw coconut fiber was collected from a coir factory and used as the base adsorbent material for fluoride adsorption studies. The material was oven-dried at 80 °C for 2 hours to remove moisture content, followed by grinding using a mechanical grinder. The powdered material was sieved using 125 BSS mesh to obtain uniform particle size suitable for adsorption studies. Due to formation of fine powder during grinding and sieving, lower yield was obtained, and a total of 112 g of processed material was collected after preparation.

Surface functionalization was carried out using Glycidyl Trimethyl Ammonium Chloride (GTMAC). Initially, the biomass was treated with NaOH solution overnight to remove lignin, hemicellulose, and surface impurities while activating cellulose hydroxyl groups for functionalization. Based on literature studies, 0.454 g of GTMAC per gram of biomass was used in approximately 50 mL solution. The reaction was carried out under alkaline conditions at pH 11–13 and maintained at 60 °C for 5 hours with sonication. The modified material was then washed to neutral pH and dried at 60 °C.

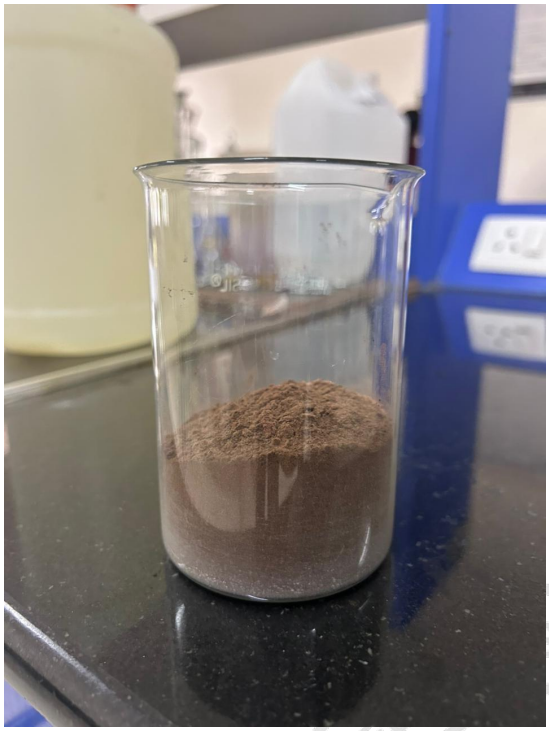


Figure 6.1: Prepared bio material

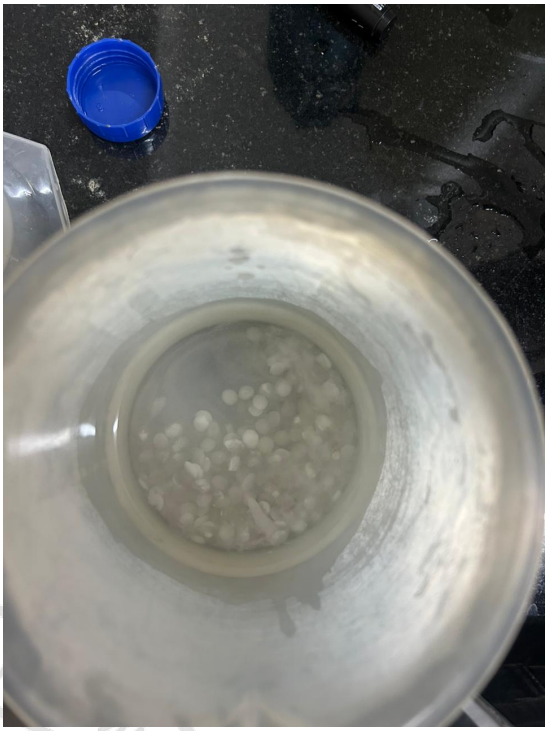


Figure 6.2: Preparation of 10% NaOH solution



Figure 6.3: Addition of biomass to 10 %(w/v) NaOH solution



Figure 6.4: Biomass in 10% (w/v)NaOH solution



Figure 6.5: Washing of soaked biomass using distilled water



Figure 6.6: GTMAC modified material kept for drying

6.2 Experimental Design and CCD Adsorption Runs

The adsorption studies were planned using Design of Experiments (DOE) through Central Composite Design (CCD). Four major adsorption parameters were selected for the study:

- pH
- Adsorbent dose
- Initial fluoride concentration
- Contact time

Approximately 30 CCD experimental runs were generated using Design-Expert software under different combinations of adsorption parameters. The generated CCD matrix provides systematic combinations of operating conditions for fluoride adsorption studies and optimization analysis.

Table 6.1: CCD experimental runs and fluoride removal response

Std	Run	A:pH Factor 1	B:Adsorbent dose g/100 mL	C:Initial Fluoride Concentration (ppm)	D:Contact time (min)	Current percentage	% Fluoride removal
8	1	7	1	70	30	17.69	74.729
13	2	2	0.1	70	180	16.07	77.043
21	3	4.5	0.55	2	105	0.409	79.550
17	4	2	0.55	36	105	7.297	79.731
19	5	4.5	0.1	36	105	10.1	71.944
3	6	2	1	2	30	0.244	87.800
22	7	4.5	0.55	70	105	19.77	71.757
4	8	7	1	2	30	0.373	81.350
2	9	7	0.1	2	30	0.353	82.350
26	10	4.5	0.55	36	105	9.527	73.536
27	11	4.5	0.55	36	105	9.753	72.908
23	12	4.5	0.55	36	30	8.235	77.125
24	13	4.5	0.55	36	180	8.311	76.914
9	14	2	0.1	2	180	0.41	79.500
20	15	4.5	1	36	105	9.766	72.872
30	16	4.5	0.55	36	105	10.6	70.556
6	17	7	0.1	70	30	17.67	74.757
18	18	7	0.55	36	105	10.61	70.528
25	19	4.5	0.55	36	105	9.823	72.714
28	20	4.5	0.55	36	105	9.975	72.292
1	21	2	0.1	2	30	0.499	75.050
11	22	2	1	2	180	0.259	87.050
5	23	2	0.1	70	30	16.12	76.971
29	24	4.5	0.55	36	105	10.02	72.167
12	25	7	1	2	180	0.574	71.300
14	26	7	0.1	70	180	13.78	80.314
16	27	7	1	70	180	19.06	72.771
7	28	2	1	70	30	12.42	82.257
10	29	7	0.1	2	180	0.438	78.100
15	30	2	1	70	180	12.57	82.043

6.3 Langmuir Adsorption Model Diagnostics

Langmuir adsorption model diagnostics were carried out during Phase 2 of the study to evaluate adsorption prediction performance. The Langmuir adsorption equation is expressed as:

$$q_e = \frac{q_{\max} b C_e}{1 + b C_e} \tag{6.1}$$

where:

- q_e = Equilibrium adsorption capacity (mg/g)
- $q_{\max}$ = Maximum monolayer adsorption capacity (mg/g)

- b = Langmuir adsorption constant related to adsorption energy
- C_e = Equilibrium fluoride concentration in solution (mg/L)

The diagnostic analysis included:

- Actual vs Predicted plot
- Residual plot
- Residual distribution histogram
- Q-Q probability plot

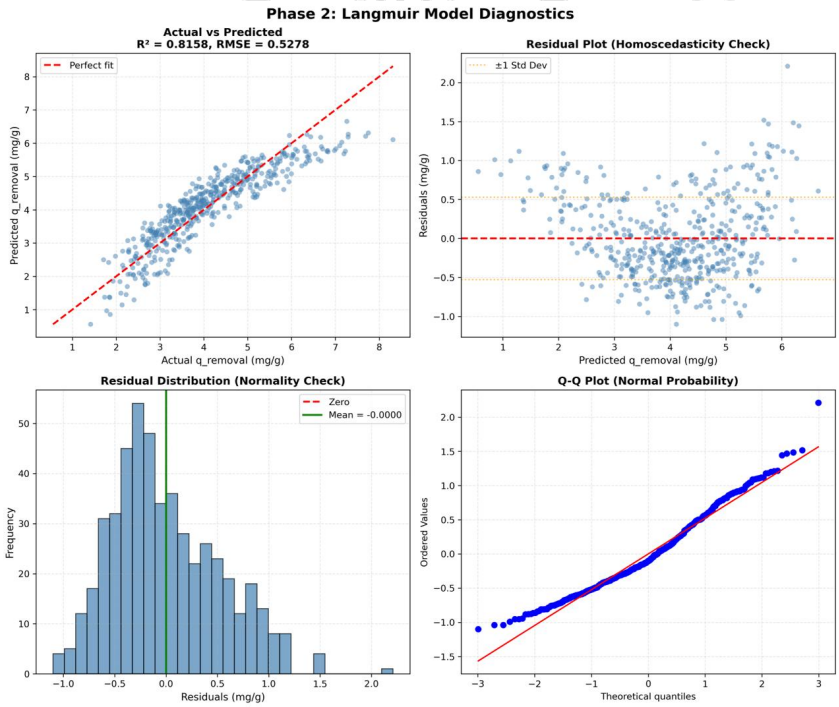


Figure 6.7: Graphs from the model built

Chapter 7

Conclusion and Future Scope

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

Development of low-cost and efficient adsorbents for fluoride removal is important for water treatment applications. In the present work, coconut fiber was selected as a biomass adsorbent and surface modification using Glycidyl Trimethyl Ammonium Chloride (GTMAC) was carried out to improve adsorption characteristics. DOE using Central Composite Design (CCD) was used for planning adsorption studies, and preliminary Langmuir adsorption model diagnostics were analyzed during Phase 2 of the work. This chapter presents the conclusions, future scope, and learning outcomes of the project.

7.1 Conclusion

The present work focused on the preparation and GTMAC functionalization of coconut fiber for fluoride adsorption studies, coupled with advanced computational modelling. Biomaterial preparation was carried out through drying, grinding, and sieving processes to obtain uniform particle size suitable for adsorption studies.

CCD experimental runs were generated using Design-Expert software by varying pH, adsorbent dose, fluoride concentration, and contact time. In the computational phase, a 500-point Latin Hypercube Sampling (LHS) design was used to simulate adsorption responses. The baseline Langmuir model showed good prediction performance with an R^2 value of 0.8158 and RMSE value of 0.5278 mg/g, confirming that fundamental chemistry captures the majority of the variance.

However, rigorous residual analysis revealed systematic, non-linear error patterns driven predominantly by pH (specifically underestimation at the optimal pH of 6.5–7). Through strategic feature engineering, 28 new features were developed, successfully demonstrating that these physical model residuals are learnable using machine learning techniques. Ultimately, the study establishes a strong foundation for a hybrid physics-ML modelling approach for environmental remediation.

7.2 Future Scope

Future work includes:

- Carrying out complete batch adsorption studies and fluoride removal experiments using the prepared GTMAC-modified coconut fiber to generate real-world experi-

mental validation data.

- Performing SEM and FTIR analysis to study surface morphology and confirm successful GTMAC functionalization.
- Executing Phase 4 of the computational workflow to train and tune advanced ML models (Random Forest, XGBoost, MLP) to correct the Langmuir residuals, with the target of achieving a hybrid model $R^2 \geq 0.94$.
- Validating the hybrid model on complex, multi-ion real groundwater samples.

7.3 Learning Outcomes of the Project

- Knowledge of GTMAC surface functionalization and biomaterial preparation.
- Understanding of Design of Experiments (DOE), CCD, and Latin Hypercube Sampling (LHS).
- Proficiency in Langmuir adsorption modelling and rigorous residual diagnostics.
- Development of skills in feature engineering and hybrid physics-informed machine learning architectures.
- Exposure to FTIR and SEM characterization techniques.
- Understanding of adsorption data analysis using both statistical software and Python-based ML pipelines.

Appendix A

Computational Workflow Scripts and Data Structures

APPENDIX A

COMPUTATIONAL WORKFLOW SCRIPTS AND DATA STRUCTURES

A.1 First Appendix (Tcl Example)

You can use `tcblisting` for creating the code snippets. The following example illustrates how one can customize the `tcblisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```

This appendix also provides pseudocode snippets and data structure examples representing the hybrid Langmuir–Machine Learning computational modelling workflow developed in Python.

A.2 Phase 1: Design of Experiments and Simulation

The following pseudocode illustrates the generation of the Latin Hypercube Sampling (LHS) matrix and the physics-based simulation of fluoride adsorption.

```
import pandas as pd
from scipy.stats import qmc
import numpy as np

# 1. Generate Latin Hypercube Sampling (LHS) Design
sampler = qmc.LatinHypercube(d=10, seed=42)
sample = sampler.random(n=500)
# Scale to factor ranges (e.g., pH: 3-9, Dose: 0.5-5)
factors = scale_to_ranges(sample)

# 2. Physics-Based Simulation
def simulate_adsorption(row):
    # Langmuir baseline capacity
    q_langmuir = q_max * KL * row['C0'] / (1 + KL * row['C0'])

    # pH effect (Gaussian bell curve centered at 6.5)
    pH_effect = np.exp(-((row['pH'] - 6.5)**2) / 4.5)

    # Kinetics, Ion Competition & Fouling penalties
    kinetics = 1 - np.exp(-k * row['Dose'] * row['Time'])
    ion_penalty = calculate_ion_competition(row['Chloride'],
                                            row['Hardness'],
                                            row['Carbonate'])
    fouling = 1 - (0.15 * row['NOM'] / 50)

    # Final response with 5% Gaussian noise
    q_rem = q_langmuir * pH_effect * kinetics * (1-ion_penalty) *
       fouling
    return q_rem + np.random.normal(0, 0.05 * q_rem)

df = pd.DataFrame(factors, columns=factor_names)
df['q_removal'] = df.apply(simulate_adsorption, axis=1)
df.to_csv('dataset_simulated_500.csv')
```

A.3 Phase 2: Langmuir Baseline Model Fitting

This snippet demonstrates the fitting of the 66-feature polynomial Langmuir baseline model and the calculation of residuals.

```
from sklearn.preprocessing import PolynomialFeatures,
   StandardScaler
from sklearn.linear_model import LinearRegression

# Load dataset
df = pd.read_csv('dataset_simulated_500.csv')
X = df[factor_names]
y_actual = df['q_removal']

# Generate 66 features (linear, squared, interactions)
poly = PolynomialFeatures(degree=2, include_bias=False)
X_poly = poly.fit_transform(X)

# Standardize and fit Ordinary Least Squares (OLS)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_poly)

model = LinearRegression()
model.fit(X_scaled, y_actual)

# Calculate predictions and residuals
df['q_predicted'] = model.predict(X_scaled)
df['residual'] = df['q_removal'] - df['q_predicted']

df.to_csv('langmuir_predictions.csv')
```

A.4 Phase 3: Feature Engineering and ML Preparation

This section illustrates the extraction of engineered features targeted at the non-linear pH residual patterns.

```
# Feature engineering targeting the Langmuir residual patterns
def engineer_features(df):
    # pH Deviation Features
    df['pH_abs_dev'] = abs(df['pH'] - 6.5)
    df['pH_dev_sq'] = (df['pH'] - 6.5)**2
    df['pH_optimal'] = df['pH'].apply(lambda x: 1 if 6 <= x <= 7
       else 0)

    # Ratios and Interactions
    df['C0_Dose_ratio'] = df['C0'] / df['Dose']
    df['total_ions'] = df['Chloride'] + (df['Hardness']/10) + \
                       df['Carbonate']
    df['carbonate_at_pH'] = df['Carbonate'] * (df['pH'] - 6.5)

    # Logarithmic transformations
    df['log_Time'] = np.log(df['Time'])

    return df

ml_data = engineer_features(df)
# Split into training (80%) and test (20%) datasets
ml_data.to_csv('ml_training_data.csv')
```

A.5 Data File Structures (CSV Snippets)

1. dataset_simulated_500.csv (Sample Data)

```
Run, pH, C0_mgL, Time_min, Dose_gL, Temp_C, ..., q_removal
1,  7.00,  70.0,      30.0,    1.00,  30.0, ..., 4.12
2,  2.00,  70.0,     180.0,    0.10,  25.0, ..., 1.45
```

```
3,  4.50,   2.0,     105.0,    0.55,  35.0, ..., 3.56
... (500 rows)
```

2. langmuir_predictions.csv (Model Output)

```
Run, pH, C0_mgL, ..., q_removal, q_predicted, residual
1,  7.00,  70.0, ...,      4.12,        3.47,     0.65
2,  2.00,  70.0, ...,      1.45,        1.97,    -0.52
... (500 rows)
```

BIBLIOGRAPHY

[1] "A comprehensive review of fluoride removal using low-cost adsorbents for environmental and industrial applications," *Sustainable Chemistry and Pharmacy*, 2025.

[2] P. Budeli *et al.*, "Fluoride removal using ce/al/fe tri-metal oxide modified diatomaceous earth," *Water Research*, 2025.

[3] M. Kamilia *et al.*, "Modification of chitosan-coated magnetic material for enhanced adsorption performance," *Journal of Environmental Chemical Engineering*, 2025.

[4] X. Yang *et al.*, "Chitosan-loaded coconut fiber for enhanced heavy metal adsorption from wastewater," *Separation and Purification Technology*, vol. 368, p. 133 105, 2025.

[5] U. Author *et al.*, "Key parameters in adsorption for effective fluoride removal," *MDPI*, 2024.

[6] S. Chauhan *et al.*, "Synthesis and investigation of gtmac modified biomass adsorbents for anionic pollutant removal," *Biomass Conversion and Biorefinery*, 2024.

[7] "Exploring key parameters in adsorption for effective fluoride removal," *Applied Sciences*, 2024.

[8] P. G. Hiremath and M. Chennabasappa, "Fluoride removal using modified rice husk biochar: Batch and column studies," *Sustainable Chemistry One World*, vol. 2, p. 100 005, 2024.

[9] A. Hussain *et al.*, "A sustainable approach for fluoride treatment using coconut fiber cellulose as an adsorbent," *Environmental Research*, vol. 244, p. 117 952, 2024.

[10] P. R. K. C. Isuranaga R.A.T., "Fluoride removal from water using activated carbon derived from biomass: Review," *Unknown Journal*, 2024.

[11] T. Ayalew, "Fluoride adsorption using thermally activated kaolin clay and limestone," *Environmental Technology & Innovation*, 2023.

[12] S. Mukherjee *et al.*, "Insight into biosorptive uptake of fluoride by chemically activated biochar: Experimental modeling and parametric optimization," *Biomass Conversion and Biorefinery*, vol. 13, no. 18, pp. 16 753–16 764, 2023.

[13] "Magnesium modified activated carbons derived from coconut shells for the removal of fluoride from water," *Separation and Purification Technology*, 2022.

[14] S. Gogoi *et al.*, "Pei–gtmac functionalized biomass adsorbents for enhanced anion removal," *Bioresource Technology*, 2019.

[15] "Isotherm and kinetic studies of fluoride removal on activated carbon obtained from coconut shell," *Innoriginal: International Journal of Sciences*, 2019.

[16] M. S. D. V. S. D. H.-S. S. O. Talat M., "Effective removal of fluoride from water by coconut husk activated carbon in fixed bed column: Experimental and breakthrough curves analysis," *Groundwater for Sustainable Development*, 2018.

[17] World Health Organization, *Guidelines for Drinking Water Quality*. World Health Organization, 2017.

[18] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.

[19] G. K. S. F. Islamuddin R., "Removal of fluoride from drinking water by coconut husk as natural adsorbent," *International Journal of Engineering Sciences and Research Technology*, 2016.

[20] "Removal of fluoride from drinking water using aluminum hydroxide coated activated carbon prepared from bark of morinda tinctoria," *Applied Water Science*, 2016.

[21] L. Chen *et al.*, "Gtmac quaternization of chitosan in ionic liquid for improved adsorption," *Carbohydrate Polymers*, 2014.

[22] . M. R. Said M., "Defluoridation of water supplies using coconut shells activated carbon: Batch studies," *International Journal of Science and Research*, 2014.

[23] R. S. M. S. D. A. Sivasankar V., "Shungite — a carbonaceous natural mineral for the defluoridation of water," *Chemical Engineering Journal*, 2013.

[24] E. R. A. E. B. H. B.-Z. M. E. H. R. T. A. Mourabet M., "Removal of fluoride from aqueous solution by adsorption on hydroxyapatite," *Desalination and Water Treatment*, 2012.

[25] K. E. S. M. Bhatnagar A., "Fluoride removal from water by adsorption — a review," *Chemical Engineering Journal*, 2011.

[26] A. S. M. B. G. D. S. P. Mohapatra M., "Review of fluoride removal from drinking water," *Journal of Environmental Management*, 2009.

[27] D. Mohan *et al.*, "Fluoride removal from water using activated alumina," *Journal of Hazardous Materials*, 2007.

[28] I. Langmuir, "The adsorption of gases on plane surfaces of glass, mica and platinum," *Journal of the American Chemical Society*, 1918.

RASHTREEYA SIKSHANA SAMITHI TRUST ® RV INSTITUTIONS